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1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

```python
m=float(input())
v=float(input())
p=m*v
print(f"{p:0.2f}kgm/s")
```

1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:

$1 \le n \le 999$

Input Format:

- The input consists of a single integer n.

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

condition...

```python
import math
n=int(input())
if n>=1 and n<=9:
    print(n*n)
elif n>=10 and n<=99:
    print(f"{math.sqrt(n):0.2f}")
elif n>=100 and n<=999:
    print(f"{math.cbrt(n):0.2f}")
else :
    print("Invalid")
```

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Terminal Test cases

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases +

noOfDay...

```python
from datetime import date
date1_str = input()
date2_str = input()

year1, month1, day1 = map(int, date1_str.split('-'))
year2, month2, day2 = map(int, date2_str.split('-'))

d1 = date(year1, month1, day1)
d2 = date(year2, month2, day2)

delta = d2 - d1
print(delta.days)
```

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
num=int(input())
reversed_num=int(str(num)[::-1])
print(reversed_num)
```

Terminal | Test cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:
- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:
- Print the multiplication table for the given number in the format.

```
<number> x <multiplier> = <result>
```

Note:
- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
n=int(input())
for i in range (1,11):
 print(f"{n} x {i} = {n*i}")
```

Terminal | Test cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases +

passorFa... Submit

```python
n = int(input())
marks = list(map(int, input().split()))

# Check fail condition
failed = False
for m in marks:
    if m < 40:
        failed = True
        break

if failed:
    print("Fail")
else:
    avg = sum(marks) / n
    print(f"Aggregate Percentage: {avg:.2f}")

    if avg > 75:
        print("Grade: Distinction")
    elif avg >= 60:
        print("Grade: First Division")
    elif avg >= 50:
        print("Grade: Second Division")
    elif avg >= 40:
        print("Grade: Third Division")
```

Terminal Test cases

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1.2.2. Fibonacci Series

05:55

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:
- A single integer n representing the number of terms to generate.

Output Format:
- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

fibonacci....

Submit

```python
# def fibonacci(n):

#   if n == 1:
#       return 0
#   elif n == 2:
#       return 1
#   else:



#       return fibonacci(n-1) + fibonacci(n-2)


# n = int(input())
# for i in range(1, n + 1):
#   print(fibonacci(i), end=" ")
# #def fibonacci(n):


# #     return fibonacci(   )



# n = int(input())
# # for i in range(1, n + 1):
#   print(fibonacci(i), end=" ")
def fibonacci(n):
    if n == 1:
        return 0
    elif n == 2:
        return 1
    else:
```

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```python
n = int(input())

for i in range(1, n + 1):
    print("* " * i)
```

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```python
# Input number of rows
n = int(input())

# Loop through rows
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

Terminal Test cases

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